🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 · STANDARD 6.DS.6D

Choose Appropriate Measures

Lesson 2-10 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can choose the best measure of center for a data set based on its shape.
Language Objective: I can explain my choice using the words mean, median, outlier, and skewed.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Mean | Median | Outlier | Skewed | Data distribution | Variability

👆 *Tap any word to see its meaning and a picture.*

1. The average of a data set is the ________.

2. The middle value when data is in order, often best when there is an outlier, is the ________.

3. A value far from the rest of the data, which can pull the mean, is an ________.

4. Data that is bunched on one side with a tail on the other is ________.

5. The way data values are spread out is the ________.

6. How much the data values differ is the ________.

Write About the Math

Is $85 a fair way to describe a 'typical' ticket price? What measure should the reporter use?

¿$85 es una manera justa de describir un precio de boleto «típico»? ¿Qué medida debería usar el reportero?

Say your answer to a partner first. Then write it, using at least two of these words:

- [ ] **Mean** *Media*
- [ ] **Median** *Mediana*
- [ ] **Outlier** *Valor atípico*
- [ ] **Skewed** *Sesgado*
- [ ] **Data distribution** *Distribución de datos*

Model: *Data with an outlier or a skewed shape is better summarized by the median.* — A strong answer says the presence of an outlier or skew signals the median, while symmetric data with no outliers fits the mean.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The reporter used the ____, which is ____. A better measure would be the ____ (about ____) because ____. The courtside seat at $260 is an ____ that pulls the mean ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Mean
2. **Notes 2:** Median
3. **Notes 3:** Outlier
4. **Notes 4:** Skewed
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Appropriate Measures is defaulting to the mean just because "it uses all the data," without first checking whether an outlier or skewed shape is present. For example, given a player's points per game 14, 16, 15, 17, 16, 58, a student calculates the mean (136 ÷ 6 ≈ 22.7) and reports that as the typical game — but 58 is far from the cluster of 14–17, so it pulls the mean above 5 of the 6 games. The median (16) actually represents a typical game here. Before choosing a measure, always scan the data for an outlier or a long tail first: only use the mean once you've confirmed the data has neither.*
8. **Try It 1:** A. Mode, because it names the value that appears most — *The mode is the value that shows up most often. Here size M appears 4 times, so the mode (M) tells the manager what to order most.*
9. **Try It 2:** A. Range, because it is the highest minus the lowest value — *Range = max − min = 18 − 6 = 12. Because it uses only the two most extreme values, one outlier (18) changes the range a lot.*